PROGRAM REVIEW · Q4 2026

Scaling LLM evaluation from one channel to every channel

Automated judging now covers 86% of production traffic. Here is what it found, and what we need next.

AI Engineering · October 2026

Automated evaluation now covers most production traffic at a fraction of the cost

86%

of production conversations scored daily

+61 pts since Q1

0.82

agreement with expert human raters (Cohen's κ)

+0.09 vs. Q2

$0.004

cost per evaluated conversation

-71% vs. manual QA

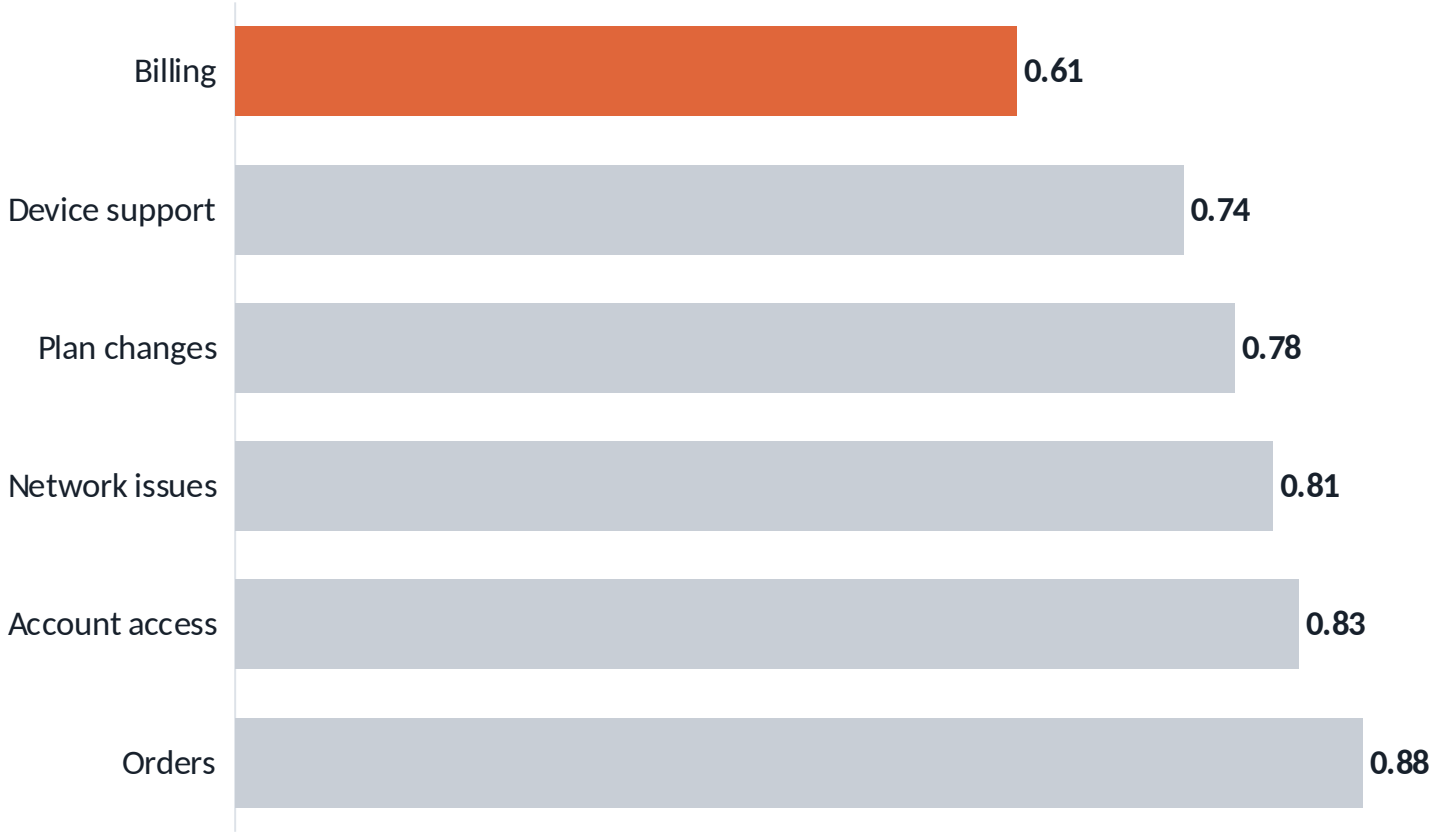
The judge is now trusted enough to gate releases, not just report on them.

01

What the judge found

Quality signals from six channels over the last 90 days

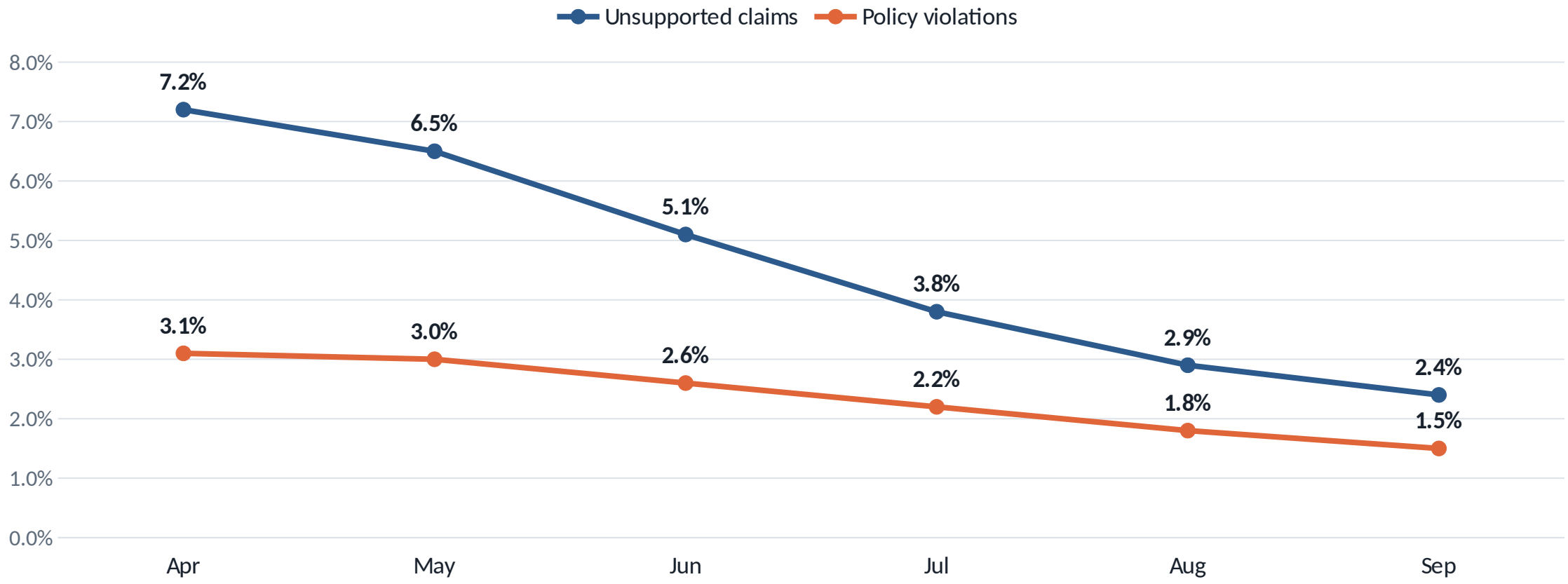
Resolution quality varies most in billing, where fixes would pay back fastest



Billing scores 0.61 — 13 points below the next-worst channel. Most failures trace to stale policy content in retrieval.

Source: Judge scores, Jul–Sep 2026 (illustrative)

Hallucination rate fell by two-thirds after retrieval and prompt fixes



Source: Share of judged conversations flagged (illustrative)

Release gating would have stopped three regressions before customers saw them

3 of 3

Q3 regressions caught in replay before rollout

We replayed every Q3 release against the judge. Each regression that reached customers would have failed the gate on at least two metrics.

Moving from sampled manual QA to continuous automated evaluation

Before

- 2% of conversations sampled by hand
- Findings arrive two weeks late
- No signal at release time

Now

- 86% scored automatically every day
- Dashboards refresh hourly
- Releases gated on judge metrics

Every channel clears the agreement bar; billing and device support need rubric work

Channel	Coverage	Judge–human κ	Resolution	Status
Billing	**91%**	**0.79**	**0.61**	**Rubric revision**
Device support	88%	0.77	0.74	Rubric revision
Plan changes	85%	0.84	0.78	Gating live
Network issues	83%	0.85	0.81	Gating live
Account access	80%	0.86	0.83	Gating live
Orders	79%	0.88	0.88	Gating live

Source: Illustrative data

02

What we need next

Three investments turn evaluation into a release control

- **Fix billing retrieval** before adding more judge coverage
  - Refresh policy corpus weekly
  - Add citation-required prompting
- **Calibrate judge scores** so thresholds mean the same thing on every channel
  - Platt scaling per channel against expert labels
- **Make the gate mandatory** for all LLM releases from January

Ask: approve two engineers for Q1 and the January gating policy.

Roll-out plan: gating mandatory for every channel by end of Q1

NOV

Billing retrieval fix

Corpus refresh and citation prompting in production

DEC

Per-channel calibration

Thresholds agreed with channel owners

JAN

Gating mandatory

All LLM releases pass replay evaluation

MAR

Full coverage

Remaining 14% of traffic onboarded



For the first time we found out about a quality problem before our customers did.

Channel owner, Device Support

Decisions requested today

1. Approve two engineers for Q1 calibration and retrieval work
2. Endorse mandatory release gating from January
3. Nominate one owner per channel for rubric sign-off

AI Engineering · ai-eng@example.com